... and finally smaller

Once it's hot enough, a red giant starts to burn a new fuel called helium. That pushes the outer layers of the star further out. The star then begins to lose these layers as a nebula, and eventually emerges as a small, white dwarf star.

All living things are made from stardust.

The Earth

Going out with a bang

Some giant stars end their lives with a huge explosion, called a supernova. Sometimes the centre will survive as a black hole or neutron star.

A giant star pictured before it exploded and formed the supernova 1987a.

1987a was the brightest supernova in the Earth's skies for almost four centuries.

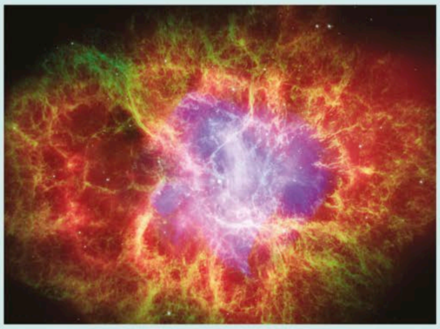
A lighthouse in space

Some neutron stars send out radio waves that sweep around as the star spins. Astronomers can pick up these signals. These neutron stars are called pulsars.

Images of dying stars

This is the young Hourglass Nebula (MyCn18) around a dying star.

Here's an example of a butterfly nebula showing its supersonic "exhaust".



A star's spectacular death in the constellation Taurus, which was observed as supernova 1054.

Its fuel will last another five billion years.